Drive systems & mechanisms

ENGINEERING WITH MR KELLY

National 5 · Mechanisms & structures · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name the four types of motion and give examples
- ☐ calculate velocity ratio, output speed and torque in gear trains
- ☐ describe the effect of an idler gear
- ☐ design a compound gear train to give a required speed
- ☐ describe ways to reduce friction and wear

I DO — Worked example

Read how a full-mark answer is built

An idler gear B is placed between gears A and C.

1 State the effect on the output speed if the number of teeth on idler B is reduced. (1 mark)

Full-mark answer each ✓ is one mark

There is no effect on the output speed; an idler only changes the direction of rotation ✓.

Why answers lose marks

"It reverses the direction of gear C."

0 marks: the question asks about speed.

WE DO — Guided practice

Use the starters and word bank to help you

Moving parts in a drive system wear quickly.

2 Describe one way to reduce wear. (1 mark)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

bearings · friction · self-lubricating · reduce · grease · harder · lubricate · materials

Tip: 1 marks = 1 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A pneumatic piston moving in and out shows...
☐ Rotary motion ☐ Oscillating motion ☐ Reciprocating motion ☐ Linear only

b) A 20-tooth driver turns a 60-tooth gear. The VR is...
☐ 1:3 ☐ 3:1 ☐ 20:1 ☐ 60:1

c) A 1500 rev/min, 52-tooth gear drives a 12-tooth gear. Output speed is about...
☐ 346 rev/min ☐ 6500 rev/min ☐ 1500 rev/min ☐ 780 rev/min

d) Reducing the teeth on an idler gear makes the output...
☐ Faster ☐ Slower ☐ No different ☐ Reverse

4 Key terms: write the letter of the matching description next to each term.

1. **Reciprocating motion**	____	**A** Backwards and forwards in a straight line.
2. **Oscillating motion**	____	**B** Two gears on one shaft; total VR = VR1 × VR2.
3. **Velocity ratio**	____	**C** Speed of input ÷ speed of output = driven teeth ÷ driver teeth.
4. **Gear speed method**	____	**D** Swinging backwards and forwards in an arc.
5. **Idler gear**	____	**E** Input speed × input teeth = output speed × output teeth.
6. **Compound gear train**	____	**F** Makes driver and driven turn the same way; doesn't change the VR.

A 20-tooth driver gear meshes with a 60-tooth driven gear. The motor turns at 1500 rpm.

5 Calculate the velocity ratio and the output speed. (3 marks)

A compound gear train: A (15 teeth) drives B (45 teeth); C (10 teeth, on the same shaft as B) drives D (50 teeth). Input 3000 rpm.

6 Calculate the output speed. (3 marks)

Machines contain several types of motion.

7 Name the four types of motion and give an example of each. (4 marks)

Check and reflect

Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?